

Models Evaluated

Model	Open-Sourced
IndicWhisper-Tamil(whisper-medium-ta_alldata_multigpu)	Yes
IndicWhisper-Hindi(whisper-medium-hi_alldata_multigpu)	
IndicWhisper-Telugu(whisper-medium-te_alldata_multigpu)	
IndicWhisper-Marathi(whisper-medium-mr_alldata_multigpu)	
IndicWhisper-Bengali(whisper-medium-be_alldata_multigpu)	
Sarvamai/saaras v3	No
Gnani Prisma v2.5	No

Test Dataset

Dataset	Languages	Description	Open Source	Approximate Size
Kathbath	12+ Indian languages	Crowdsourced read speech dataset for ASR research	Yes	~7k hrs
IndicVoices	22 Indian languages	Large-scale crowdsourced speech corpus covering diverse accents and domains	Yes	~17k hrs

Evaluation metrics

1)Word Error Rate

Measures the percentage of incorrectly recognized words.

$$\text{WER} = (S + D + I)/N$$

Where:

- **S** = Word Substitutions
- **D** = Word Deletions
- **I** = Word Insertions
- **N** = Total words in the reference transcript

Lower WER indicates better ASR performance. WER is the most widely used metric for evaluating ASR systems

2)Character Error Rate

Measures errors at the character level instead of the word level.

$$\text{CER} = (S + D + I)/N$$

Where:

- **S** = Character substitutions
- **D** = Character deletions
- **I** = Character insertions
- **N** = Total characters in the reference transcript

Lower CER indicates better performance. CER is especially useful for languages with rich morphology when evaluating multilingual ASR systems.

Reported Benchmark Results(WER)

Table 4: Performance of Publicly available models on the Vistaar Benchmark

Datasets	bn	gu	hi	kn	ml	mr	or	pa	sa	ta	te	ur	avg
Google STT													
Kathbath	20.6	26.0	14.3	27.9	52.6	26.6	105.7	25.5	104.2	31.1	34.3	17.6	40.5
Kathbath Hard	22.4	28.4	16.6	31.4	53.6	28.4	105.4	29.2	104.4	32.8	35.9	20.2	42.4
CommonVoice	26.1	-	20.8	-	54.0	27.5	101.7	29.2	-	42.1	-	26.9	41.0
FLEURS	27.3	34.7	19.4	33.8	41.0	33.2	100.8	44.0	-	36.6	40.1	28.4	39.9
IndicTTS	37.0	33.6	18.3	32.8	38.3	21.8	104.7	-	-	31.3	55.4	-	41.5
MUCS	-	59.7	17.8	-	-	21.4	103.5	-	-	27.8	46.4	-	46.1
Gramvaani	-	-	59.9	-	-	-	-	-	-	-	-	-	59.9
Average	26.7	36.5	23.9	31.5	47.9	26.5	103.6	32.0	104.3	33.6	42.4	23.3	44.3
Azure STT													
Kathbath	16.7	22.4	13.6	23.6	43.3	20.9	NA	NA	NA	28.1	22.8	NA	-
Kathbath Hard	18.4	26.7	14.9	25.9	45.7	-	NA	NA	NA	30.0	24.6	NA	-
CommonVoice	15.0	-	14.6	-	47.5	19.6	NA	NA	NA	39.9	-	NA	-
FLEURS	29.9	34.8	24.3	31.0	39.2	37.2	NA	NA	NA	34.9	32.3	NA	-
IndicTTS	21.2	50.5	15.2	25.3	33.5	31.6	NA	NA	NA	31.2	61.1	NA	-
MUCS	-	23.7	15.1	-	-	27.5	NA	NA	NA	25.1	16.1	NA	-
Gramvaani	-	-	42.3	-	-	-	NA	NA	NA	-	-	NA	-
Average	20.2	31.6	20.0	26.4	41.8	27.4	NA	NA	NA	31.5	31.4	NA	-
IndicWav2Vec													
Kathbath	14.3	17.4	12.2	20.8	37.8	20.4	22.2	15.1	37.0	27.4	26.9	16.0	22.3
Kathbath Hard	18.1	21.5	16.2	26.4	42.8	24.0	28.3	18.2	43.3	33.0	31.2	19.4	26.9
FLEURS	28.4	22.7	18.3	22.0	40.4	25.1	40.5	34.8	-	37.7	30.4	32.2	30.2
CommonVoice	31.3	-	20.2	-	67.0	27.0	37.9	19.7	-	42.7	-	32.7	34.8
IndicTTS	28.1	22.7	15.0	20.1	40.5	18.6	24.4	-	-	28.0	32.3	-	25.5
MUCS	-	32.6	22.9	-	-	34.7	30.1	-	-	37.7	30.4	-	31.4
Gramvaani	-	-	42.1	-	-	-	-	-	-	-	-	-	42.1
Average	24.0	23.4	21.0	22.3	45.7	25.0	30.6	21.9	40.2	34.4	30.2	25.1	28.7
IndicWhisper													
Kathbath	16.6	17.8	10.1	19.3	34.8	19.9	24.7	16.9	45.6	24.2	25.0	11.9	22.2
Kathbath Hard	19.4	20.6	11.9	22.2	38.4	22.1	29.1	19.7	50.5	27.5	27.8	14.7	25.3
CommonVoice	24.7	-	14.2	-	44.5	22.8	35.2	22.4	-	29.2	-	31.7	28.1
FLEURS	20.9	23.5	10.5	18.6	22.6	20.5	32.9	23.1	-	25.2	25.4	19.2	22.0
IndicTTS	18.8	19.1	8.1	13.2	21.4	11.4	15.0	-	-	17.2	33.8	-	17.6
MUCS	-	33.2	17.1	-	-	12.8	27.5	-	-	28.3	32.1	-	25.2
Gramvaani	-	-	24.9	-	-	-	-	-	-	-	-	-	24.9
Average	20.1	22.8	13.8	18.3	32.3	18.2	27.4	20.5	48.0	25.3	28.8	19.4	24.6

Per Language WER/CER Results

Language	Model	Source Type	WER(%)	CER(%)	Test Set	Test Set Size	Rank
Hindi	Vaani-FastConformer-Hindi	Open	15.11	7.09	IndicVoices-Hindi	5530	1
	IndicConformer-600m-multilingual	Open	15.3(RNNT) 16.6(CTC)	7.2(RNNT) 7.3(CTC)			2
	IndicWav2Vec-Hindi	Open	38.6	22.5			3
	Gnani Prisma v2.5	Closed	8.78	3.26	Kathbath-Hindi	1929	1
	IndicWhisper-Hindi	Open	10.23	3.65			2
	Sarvam Saaras v3	Closed	14.78	4.59			3
Bengali	IndicConformer-600m-multilingual	Open	13.49(RNNT) 14.23(CTC)	5.36(RNNT) 5.41(CTC)	IndicVoices-Bengali	3906	1
	IndicWav2Vec-Bengali	Open	46.7	21.2			2
	IndicWhisper-Bengali	Open	19.22	5.51	Kathbath-Bengali	1783	1
	Sarvam Saaras v3	Closed	21.21	5.07			2
	Gnani Prisma v2.5	Closed	13.94	4.79			
Marathi	IndicConformer-600m-multilingual	Open	14.9(RNNT) 16(CTC)	5.61(RNNT) 5.69(CTC)	IndicVoices-Marathi	3552	1
	IndicWhisper-Marathi	Open	20.25	6.52	Kathbath-Marathi	1631	1
	Sarvam Saaras v3	Closed	20.46	5.96			2
	Gnani Prisma v2.5	Closed	16.89	5.72			
Telugu	IndicConformer-600m-multilingual	Open	26.28(RNNT) 27.86(CTC)	8.42(RNNT) 8.53(CTC)	IndicVoices-Telugu	3295	1
	Vaani-FastConformer-Telugu	Open	27.31	8.77			2
	ARTPARK-IISc/whisper-medium-vaani-telugu	Open	61.69	25.69			3
	Sarvam Saaras v3	Closed	24.8	3.7	Kathbath-Telugu	1492	1
	Gnani Prisma v2.5	Closed	25.53	7.55			2
	IndicWhisper-Telugu	Open	36.9	12.9			3
Tamil	IndicConformer-600m-multilingual	Open	30.7(RNNT) 34.7(CTC)	9.1(RNNT) 10.1(CTC)	IndicVoices-Telugu	5276	1

	IndicWhisper-Tamil	Open	24.9	4.47	Kathbath-Tamil	1642	1
	Sarvam Saaras v3	Closed	28.65	4.74			2
	Gnani Prisma v2.5	Closed	24.96	5.51			

The Problem with Standard WER/CER for Indian Languages

Standard metrics like WER and CER were developed primarily for English. They work well when every word has a single accepted spelling, when languages don't mix mid-sentence, and when the gap between formal and colloquial usage is narrow. Indian languages don't fit that description.

Sarvam identified **6 specific failure classes** where WER/CER give wrong verdicts:

1. **Colloquial variants** — A Tamil ASR that outputs the colloquial form *avunga* instead of the formal *avarga* gets WER=80%, even though a native speaker hears it as a perfect transcription.
2. **Code-mixing script mismatch** — Writing डॉक्टर (Devanagari) vs *doctor* (Latin) are both correct, but WER flags it as a substitution error.
3. **Short helper words** — A repeated word like नहीं नहीं instead of नहीं gives WER=300%, making the model look catastrophically wrong.
4. **Agglutinative languages** — In Malayalam/Telugu, a minor suffix change on a long compound word creates a huge CER penalty despite no meaningful difference.
5. **Numeric format variability** — पांच सौ, 500, and ५०० all mean the same thing but WER treats them as completely unrelated tokens.
6. **Semantic drift blindness** — Inserting नहीं (not) reverses the entire meaning but WER only gives a ~14% penalty.

The Fix: LLM-WER and LLM-CER

LLM-WER and LLM-CER address these failure classes by replacing rigid string-matching with a semantic and phonetic judgement made by an LLM. Segments that are meaning-equivalent or phonetically similar are treated as correct. Only genuinely wrong segments count as errors.

It works in **3 steps**:

1. Run standard WER/CER alignment to find mismatched segments
2. Ask an LLM: *are these semantically equivalent? are they phonetically similar?*
3. Only count segments the LLM confirms as genuinely wrong

Key insight - WER and CER measure the distance between two strings. The LLM-based metrics measure the distance between two meanings. For Indian languages, those two distances can be very different. Combining both perspectives gives the most accurate picture of ASR quality.

CONCLUSION

- **IndicConformer-600M-Multilingual with RNNT decoding is the strongest open-source model across all five languages**, achieving the best WER in Bengali (13.49%), Marathi (14.9%), Hindi (15.3%), Telugu (26.28%), and Tamil (30.7%). The RNNT decoder consistently outperforms CTC across every language and metric, with the gap being most notable in Tamil (30.7% vs 34.7%) and Telugu (26.28% vs 27.86%).
- **Among commercial APIs, Gnani Prisma v2.5 outperforms Sarvam Saaras v3 on Hindi** (8.78% vs 14.78% WER) and Telugu (25.53% vs 24.8% WER), while Saaras v3 has a slight edge on Telugu CER (3.7% vs 7.55%). Notably, Gnani Hindi (8.78% WER) is the single best result across all models and languages in this benchmark, outperforming even IndicConformer on Kathbath-Hindi.
- **Vaani-FastConformer-Hindi** is the sole open-source model that challenges IndicConformer, edging it on Hindi (15.11% vs 15.3% RNNT), making it the preferred choice for Hindi-only deployments. In contrast, **IndicWav2Vec models lag far behind**, scoring 38.6% WER on Hindi and 46.7% on Bengali compared to sub-17% and sub-14% from IndicConformer, making them unsuitable for production use.
- **Tamil remains the hardest language in this benchmark**, with the best result at 30.7% WER (IndicConformer RNNT), nearly twice the error rate of Bengali (13.49%), suggesting limited training data coverage or greater acoustic complexity. **Bengali is the easiest**, with consistently strong results across all models.
- Overall, for multilingual Indic ASR, **IndicConformer-600M RNNT is the recommended open-source baseline**, while **Gnani Prisma v2.5 is the best-performing commercial option** tested in this benchmark. Future work should prioritize improving Tamil and Telugu coverage, evaluating on IndicVoices for commercial APIs, and incorporating **LLM-WER** for semantically-aware evaluation of Indic transcriptions.